

AMIRHOSEIN CHAHE

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RESEARCH INTERESTS

World models and model-based planning, vision-language-action (VLA) policies, closed-loop training and evaluation, safe and robust autonomy for driving and navigation, efficient embodied inference

EDUCATION

Drexel University Ph.D., Electrical Engineering (expected 2028) Advisor: Prof. Lifeng Zhou	<i>2023 - Present</i>
Drexel University M.Sc., Electrical Engineering	<i>2023 - 2025</i> GPA: 3.9
University of Tehran B.Sc., Mechanical Engineering	<i>2017 - 2021</i> GPA: 3.9

RESEARCH EXPERIENCE

Research Intern , Honda Research Institute USA, Inc. Distilled a 10B driving model into a 4B one-step (NFE=1) vision-language policy for the NVIDIA AlpaSim Closed-Loop Challenge, with closed-loop safety evaluation	<i>Jun 2026 - Sep 2026</i>
Research Intern , Honda Research Institute USA, Inc. - Built a pipeline that uses Planning KL-divergence (PKL) to rank occluded agents on nuScenes by their impact on the ego plan, producing a new benchmark of planning-critical occlusions with structured VLM annotations (IROS 2026) - Benchmarked 17 VLMs and fine-tuned 14; PKL-guided data selection improved accuracy by $\sim 30\%$ over random selection, and a fine-tuned 3B model outperformed a $24\times$ larger zero-shot model	<i>Jun 2025 - Sep 2025</i>
Research Assistant , Drexel University - Developed latent world models with autoregressive rollout training and a Monotone Cost Ranking loss that shapes the planning cost for sampling-based MPC; state of the art on GNM image-goal navigation ($2.7\times$ lower orientation error than DINO-WM; beats NWM, OmniVLA, NoMaD) - Deployed the world-model planner zero-shot on a Clearpath Husky with onboard NVIDIA Jetson AGX Orin inference, reaching the goal in 5/5 indoor/outdoor trials vs. 2/5 for OmniVLA - Proposed PiJEPA, which warm-starts MPPI over a JEPA world model with a fine-tuned Octo VLA policy prior for language-conditioned navigation; best trajectory accuracy on CAST with DINOv2 and V-JEPA-2 encoders (CVPR Workshops 2026) - Demonstrated the first real-world dynamic adversarial patch attack, displayed on a screen mounted on a moving vehicle, to manipulate an autonomous vehicle's decisions (RSS 2024) - Built efficient, deployable language models for driving: reasoning-enhanced small VLMs for driving VQA (CVPR Workshops 2025) and LLM-guided open-vocabulary 3D Gaussian scene querying (Best Paper, WACV Workshops 2025)	<i>Apr 2023 - Present</i>
Researcher , Federal Aviation Administration ASSURE A71 Project	<i>2024 - 2025</i>

- Developed safety assessment algorithms and Monte Carlo simulations for uncertainty analysis in autonomous systems

PUBLICATIONS

Conference Papers

A. Chahe, T. Naes, J. D'sa, F. M. Tariq, S. Bae, L. Zhou, and D. Isele. "What's Hidden Matters: Identifying Planning-Critical Occluded Agents using Vision-Language Models," *IROS 2026* ([paper](#))

A. Chahe, C. Wang, A. Jeyapratap, K. Xu, and L. Zhou. "Dynamic Adversarial Attacks on Autonomous Driving Systems," *RSS 2024* ([paper](#))

A. Ardekani, **A. Chahe**, and M. Hairi Yazdi. "Combining Deep Learning and Game Theory for Path Planning in Autonomous Racing Cars," *ICRoM 2022* ([paper](#))

Workshop Papers

A. Chahe and L. Zhou. "Policy-Guided World Model Planning for Language-Conditioned Visual Navigation," *CVPR Workshops 2026* ([paper](#), [code](#))

A. Chahe and L. Zhou. "ReasonDrive: Efficient Visual Question Answering for Autonomous Vehicles with Reasoning-Enhanced Small Vision-Language Models," *CVPR Workshops 2025* ([paper](#), [code](#))

A. Chahe and L. Zhou. "Query3D: LLM-Powered Open-Vocabulary Scene Segmentation with Language Embedded 3D Gaussians," *WACV Workshops 2025*, **Best Paper Award** ([paper](#))

Preprints

A. Chahe, S. Cai, and L. Zhou. "Latent World Models with Monotone Planning Costs for Image-Goal Navigation," arXiv preprint, 2026 ([paper](#))

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

AI/ML Frameworks: PyTorch, Hugging Face Transformers, DeepSpeed, LoRA fine-tuning (ms-swift), TensorFlow

Methods: Latent (JEPA) World Models, Sampling-Based MPC (CEM, MPPI), Diffusion Policies, Policy Distillation, VLM Fine-Tuning, 3D Gaussian Splatting, Adversarial Robustness

GPU Computing: Multi-GPU training on NVIDIA H100/H200 clusters (bf16 mixed precision), onboard inference on NVIDIA Jetson AGX Orin

Robotics and Simulation: ROS 2 (Nav2), Clearpath Husky, AlpaSim, CARLA, Gazebo, Simulink, Docker

Other: Git, Linux, LaTeX

AWARDS AND ACHIEVEMENTS

Best Paper Award, LLVM-AD Workshop, WACV 2025 2025

Awarded for "Query3D: LLM-Powered Open-Vocabulary Scene Segmentation"

Seaman Award, College of Engineering, Drexel University Feb 2024

Awarded for outstanding academic and research achievements

Full Teaching Fellowship, Drexel University Feb 2024

Top Class Graduate, University of Tehran Sep 2021

Ranked among the top 15% of graduate students